Data Cleaning & Preparation: Change Log

DecodeLabs Industrial Training Kit | Data Analytics | Project 1 | Batch 2026
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1. Dataset overview

Item	Detail
Source file	Dataset_for_Data_Analytics.xlsx
Records (before / after)	1,200 / 1,200
Columns	14: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice
Date range	2023-01-01 to 2025-06-30
Tools	Python (pandas, openpyxl, reportlab); reproducible script included: data_cleaning_script.py

2. Audit summary (raw data)

Check	Result	Action
Missing / null values	309 (all in CouponCode)	Filled with 'No Coupon'
Duplicate OrderID / TrackingNumber / full rows	0 / 0 / 0	None needed
Invalid or text dates	0	None needed; ISO format applied
Whitespace, case or spelling variants	0	None needed
ID format violations	0	None needed
TotalPrice != Quantity x UnitPrice	0	None needed

3. Change log

Change ID	Description	Impact	Status
CR001	Null scan on all 14 columns: 309 blank values found, all in CouponCode (25.75% of rows); the other 13 columns have 0 nulls. A blank coupon means no coupon was used, so it was filled with the category label 'No Coupon' (a coupon code cannot be averaged, so mean/median imputation does not apply and mode would invent a false coupon).	Preserved 309 records (listwise deletion would have lost 25.75% of the data). CouponCode nulls after cleaning: 0.	**Resolved**
CR002	Duplicate audit: full-row duplicates = 0, duplicate OrderID = 0, duplicate TrackingNumber = 0, near-duplicates (all columns except OrderID and TrackingNumber) = 0. 11 CustomerIDs appear more than once, but no customer has the same date and product twice (same-customer/date/product duplicates = 0), so these are repeat orders and were kept.	0 rows removed. All 1,200 records retained; OrderID is unique.	**Verified, no action needed**
CR003	Date audit: 1,200 of 1,200 values are true date values, 0 text or invalid dates, 0 with a time component. Range 2023-01-01 to 2025-06-30. Explicit ISO 8601 cell format YYYY-MM-DD applied to the whole column.	0 dates changed, 0 incorrectly formatted dates remain.	**Resolved**
CR004	Numeric precision: UnitPrice and TotalPrice set to 2 decimal places (9 UnitPrice and 27 TotalPrice values were stored as whole numbers, so they displayed inconsistently). Quantity and ItemsInCart set to whole numbers.	Display consistency only; 0 stored values changed.	**Resolved**

Change ID	Description	Impact	Status
CR005	Text standardisation: trimmed spaces, collapsed double spaces and applied Proper Case to Product, PaymentMethod, OrderStatus and ReferralSource. Unique values: Product 7, PaymentMethod 5, OrderStatus 5, ReferralSource 5, CouponCode 4 (incl. 'No Coupon'). Coupon codes are intentionally kept in upper case.	0 values changed; no spelling or case variants were present.	**Verified, no action needed**
CR006	Integrity rules: ID formats (OrderID ORD+6 digits, CustomerID C+5 digits, TrackingNumber TRK+8 digits) = 0 violations; TotalPrice = Quantity x UnitPrice = 0 mismatches; ItemsInCart >= Quantity = 0 violations; zero or negative numeric values = 0.	No corrections required; business rules hold on every row.	**Verified, no action needed**
CR007	Added a Validation sheet with live Excel formulas (duplicate IDs, date validity, nulls, price rule, spaces) and kept the untouched source in a Raw_Data sheet for traceability.	Gives a reviewer a one-click proof of the 0% error threshold.	**Resolved**

4. Verification gate (threshold for Project 2)

Requirement	Proof	Result
0% error rate on unique identifiers	1,200 unique OrderIDs in 1,200 rows; 0 duplicate OrderID, 0 duplicate TrackingNumber	**PASS**
0% error rate on date formats	1,200 of 1,200 dates valid; all stored as true dates with format YYYY-MM-DD	**PASS**
No missing values left	0 blank cells in the cleaned table	**PASS**

Assumption: a blank CouponCode is interpreted as 'no coupon used'. If the business confirms that blanks mean 'data not captured', the label can be changed to 'Unknown' with one edit in the script. The Cleaned_Dataset.xlsx file contains a Validation sheet with live formulas that re-prove every figure above.